LINKQUEST TRACKLINK 5000 SERIES USER MANUAL

Quick Links

Table of Contents

LinkQuest Inc.

TrackLink 5000 USBL Tracking System

User's Guide

Version 5000.6.2

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Table of Contents

The limited warranty is void if the product housing is opened for purpose other than change of battery pack.

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1 Introduction

Thank you for purchasing LinkQuest's TrackLink 5000 system, an USBL acoustic tracking system with integrated high-speed acoustic communication capability. This user's guide applies to TrackLink 5000LC, TrackLink 5000MA and TrackLink 5000HA systems. Before operating the system you should read through the entire user's guide.

The TrackLink system tracks up to 8 targets up to 4000 meters of range using TN5010 transponders and up to 5000 meters of range using TN5015 transponders with ship noise level equivalent to the ambient noise level at (deepwater) Sea State 6.

The main differences among the 5000LC, 5000MA and 5000HA systems are the cost and positioning accuracy. The specifications of the systems are provided in Appendix A. The selection of the transponders is in Appendix B. Customized transponders are also available upon request. Utilizing its benchmark **Broadband Acoustic Spread Spectrum** (BASS) technologies, LinkQuest provides the end users with solutions for underwater tracking and communication at sharply reduced cost and increased accuracy and robustness.

The position of the submerged targets relative to the ship is determined by a transceiver system and sent to a PC via a RS232/RS422 connection. The TrackLink Navigator software integrates the transceiver with the ship's GPS/DGPS and a compass. The targets' relative bearings are adjusted to a true bearing using the information from the compass. The actual position (latitude/longitude) of the target is then calculated based on the range, true bearing and the surface support ship's position as supplied by a GPS/DGPS receiver. The results are then graphically displayed in the software. The positioning and related data may be stored and retrieved for post processing.

The TrackLink system also has optional two-way acoustic communication capability. It provides a flexible and reliable way for users to communicate with the target while tracking it.

Main features of the TrackLink system are as follows:

- Extensive use of modern digital signal processing techniques and state-of-the art DSP processors sharply reduced cost for end users.
- **Broadband Acoustic Spread Spectrum** (BASS) technologies.
- Strong rejection to multipaths and ship noise.
- Fully integrated with LinkQuest's most advanced high-speed acoustic modems. The USBL transponder and acoustic modem share the same electronics and transducer. It significantly reduces the total system size, weight and power consumption and eliminates acoustic interference.
- Smaller and lighter transceivers allow easier and less costly installation from a ship.

- Advanced power-efficient DSP technology for the USBL intelligent transponders results in extended field operating duration.
- PC Windows tracking software to display range, bearing, depth, GPS position, acoustic communication data and other information from other sensors such as an altimeter. Interfaces directly to the transceiver. No need for a heavy proprietary deck unit.

2 Shipping Contents

The transit case typically contains one transceiver, one transponder, one 70 feet cable, one 2-meter transponder cable and a CD for TrackLink Navigator software.

Warning: The transceiver is a delicate acoustic instrument. There are multiple acoustic elements inside. Excessive force will either damage the acoustic elements or affect the calibration which will result in less accuracy during usage. The user should exercise caution during usage and transportation. The user should dismount the transceiver after usage and place the transceiver inside the transit case during transit. The user should avoid exposing the transceiver to direct sunshine.

The transceiver is placed in a specially designed transit case. It is highly recommended to use the transit case during transit. The protective transparent film is used to further protect the surface of the hydrophone head. This film should not be removed.

If the user wishes to place the transceiver in his/her own transit case, he/she needs to place in the similar way. The case needs at least 2 inches of foam on top and bottom of the transceiver. The user also needs to make a plastic film to protect the hydrophone head.

Warning: The warranty is void if the transceiver is damaged from placement in an improper transit case.

The transceiver contains shock sensors. If more than 30g of force are applied on the transceiver, the transceiver may be damaged.

Warning: The warranty is void if the shock sensor indicates more than 30g of force is applied to the transceiver during usage or transit.

3 Installation

3.1 Transceiver and Transponder Installation

CAUTION

Transmitting in air at a high power level is prohibited for the transceiver and transponder. Damage to the transducer may result from it.

The TrackLink system is designed to track targets acoustically by interrogating the target and to derive range and bearing information from the returned signal from a target. The typical usages of the system are to track ROV, AUV, tow fish and diver.

3.1.1 Transceiver Introduction and Installation

The TrackLink transceiver consists of two components, i.e. a hydrophone array and an acoustic transmitter. The acoustic transmitter is connected to the hydrophone array through a short cable and is detachable. The user does not need to install the acoustic transmitter when tracking the target using Electrical Trigger mode, i.e. tracking a responder.

The user should install the transceiver to a rigid fixture on a ship. Typical installation is from a steel/iron pole or a gate valve. The user should face the transducer (the black end of the transceiver) to the seabed and submerge the transceiver in the water. The user can also tilt the transceiver towards the target for tracking purpose. See the installation drawing for more details. The user should align the X/Y axis in the installation drawing with the X/Y axis in the TrackLink Navigator software.

The detachable acoustic transmitter is typically installed at the same level and close to the hydrophone array. However, it can be also installed certain distance away from the hydrophone array if necessary.

Warning: The transceiver is a delicate acoustic instrument. Excessive shock and vibration will either damage the acoustic elements or affect the calibration which will result in less accuracy during usage. The user should exercise caution during installation and usage to avoid excessive shock and vibration. The user should dismount the transceiver after usage and place the transceiver inside the transit case during transit.

Warning: Shock sensors is installed inside the transceiver. The warranty is void if the shock sensor indicates more than 30g of force is applied to the transceiver during usage or transit.

To achieve optimal performance, the user should install the transceiver 1 meter (3 feet) below the ship's hull when installed from a pole and flush when installed from a gate valve.

The accuracy of the absolute position depends on the accuracy of the true bearing, GPS/GPS and pitch and roll compensation. The user should choose the proper instruments and operate them correctly. Both sound speed and the bending of acoustic signal in water will impact position accuracy. A thorough study of the temperature and salinity in the area of operation will help to improve the accuracy.

Static Heading Offset and pitch/roll offset caused by installation also directly impact position accuracy. The user is advised to review the section on Calibration to properly compensate for these offsets.

The TrackLink transceiver is connected to a PC on the ship through a cable. The PC software communicates with the transceiver using an RS232 serial link. Before using the transceiver, the user should check the capability and the sufficiency of the power supply for operation.

Please refer to LinkQuest's installation drawings for installation. The X, Y directions are represented by the X, Y of the TrackLink Navigator software.

Warning: The user should minimize the shock to the transceiver system installed from the ship. The shock limits for the transceiver system is defined in the Specification A. When the ship is moving at high speeds, the user should remove the transceiver from the steel pole so that the vibration caused by the ship's high speeds will not damage the transceiver.

3.1.2 Transponder Introduction and Installation

The transponder is either powered externally or internally. Before using the transponder, the user should check to be sure that the power supply has sufficient reserves to operate the transponder for the desired service time (refer to Appendix B).

The transponder can be configured to the "Transponder", the "Responder" or the "Pinger" mode. In the "Transponder" mode, the transponder is in the standby mode while it is not in active operation. It will revert to "Wake Up" mode when commanded by the TrackLink Navigator software. The transponder will then stay in the active responding mode one hour from the last time a valid acoustic interrogation from the transceiver is received.

The transponder can also be configured to the “Responder” mode. In this mode, the transponder will be triggered by the electrical signal. Details can be found in Appendix D.

The transponder can be configured to the “Pinger” mode to output acoustic signals every second. This mode is mainly used for testing purposes.

Important: The transponder with internal batteries (Model C) is typically shipped with a power-on plug. The transponder is in complete power off state when the plug is not plugged. Therefore this power-on plug should not be plugged into the transponder during storage. The power-on plug needs to be plugged into the transponder during normal operation.

Warning: The user should plug and unplug the power-on plug by holding firmly on the top (hard) part of the plug as shown in Figure 1. Squeezing or pulling on the tail of the plug may cause breaking of the wire connections inside the plug. The damage caused may result in non-functioning of the transponder.

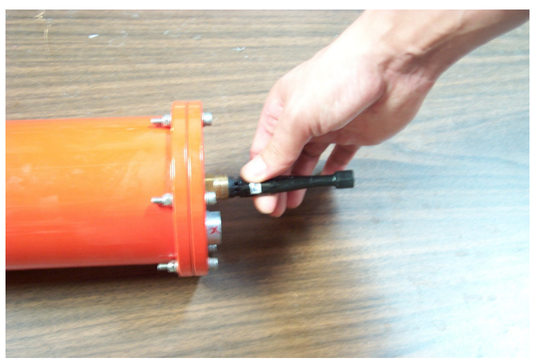


Figure 1

The user should install the transceiver and transponder properly so that the tracked target will be within the operational beamwidth of the transceiver and the transponder.

Warning: The limited warranty is void if the housing of the transceiver is opened. The limited warranty is also void if the transponder housing is opened for any purpose other than changing of batteries or the electronics of the transponder is tampered.

The product should only be used under non-corrosive environment. Non-corrosive environment may include both fresh and salt water. The product housing must be protected against abrasion that may damage the housing’s protective coatings. This product is equipped with corrosion preventive mechanism (i.e. zinc anode). If the product housing were to be electrically connected to other underwater instruments, the connection must be made such that the effect of the anti-corrosive mechanism is not diminished.

Caution: The user should make sure that the transducer on the transponder will not take any direct hit, scratch or drag. The user is advised to use a guard to protect the transducer. A guard can be typically made of thin steel rails. The diameter of the rail is recommended to be less than 5 mm. The rail should be at least 1.5 cm away from the transducer.

3.2 Tracking Software Installation

The TrackLink Navigator software displays the positions of the ship and the targets in various plots and textual displays.

IMPORTANT: If there is any TrackLink program installed on this computer, including demo and previous versions, please uninstall it first and delete the C:\tracklink directory.

IMPORTANT: Please go to "Control Panel" -> "Regional Settings" to make sure "English (United States)" is selected. It is required for the software to work properly.

Insert the software CD into CD-ROM drive. Run the setup.exe program to install the TrackLink Navigation software.

Run the software from Start->Program->tracklink. The user may also create a shortcut for the program on the desktop.

The user needs to enter the serial number twice the first time running the software. Please make sure to enter the correct number.

4 Software Operating Instructions

This section describes the functions provided by the menu bar of the TrackLink Navigation software. The user should first read through this section before proceeding to the next section on interfacing and calibration.

Warning: The TrackLink Navigator software relies on the serial communication to interface with the transceiver, GPS, compass, motion sensor/VRU and other devices. In order to minimize the impact of buffer overflow and delayed responses to serial communication, the user is advised to exit all nonessential applications running on the same computer.

IMPORTANT: The user needs to make sure the serial ports work reliably with the external devices such as GPS, compass and motion sensor. The user is advised to test the serial ports' connections to the external devices using HyperTermial first. <u>In particular when a USB/RS232 converter is used to provide serial port support, the use needs to test the reliability of this serial port.</u> There are occasions computer crashes when such a serial port is used to interface to GPS, compass or motions sensor. When it happens, the user is advised to use the original serial ports of the computer to connect to the external devices.

<u>IMPORTANT: If the RS232 communication between the computer and the transceiver is impacted by electrical noise, the user should improve the grounding conditions of the computer. The user may connect the metal shields of the transceiver cable to the ground of the computer. For notebook computers, the ground is the screw hole for the DB9 connector. The user may further improve the ground condition by connecting the computer ground to water. On a metal ship, connecting to the ship deck is equivalent to connecting to water.</u>

4.1 File

Load

This loads the history of ship and target positions. The software uses the history of positions as the starting points of the plot.

Save As

Saves the current positions, including the loaded history positions, to a file.

Save Bitmap

Saves the entire software display to a bitmap file.

Print

Prints the whole software display.

Exit

Exits the TrackLink Navigator software.

4.2 Run

Reset

Resets the transceiver.

Start

Starts both target(s) and ship tracking.

Clear

Clears all the positions in the plots. The historical positions will not be cleared. They can be saved to a user-selected file after the tracking is stopped.

Stop

Stops both target and ship tracking.

Wake Up

Wakes up a transponder with the selected address. It lasts for about 20 seconds for a selected target. This command can also wake up all targets at once by selecting "All" in the address pull down menu. There is no need to run this command if the transponder is configured to the "Responder" mode.

Warning: The user should avoid running the wake up command continuously for more than 2 minutes.

4.3 View

Choose the "Bow Reference" view or the "North Reference" view. In the "Bow Reference" view, the plot will show the target's relative positions to the ship (transceiver). The heading of the ship is displayed at the up right hand corner of the plot when a compass interface is available.

In "North Reference" view, the plot will show the target's absolute positions in longitude and latitude. The ship's bow rotates with reference to the direction "North".

The most recent 300 positions are used for the trajectory of the targets.

"Scale Down" scales down the main tracking plotting. This command is enabled only when the tracking is active.

"Scale Up" changes the TrackLink Navigator window to full screen. This command is only available when the screen is in "Scale Down" status. Please note the Maximum button on the up right corner of the window does not work at this time.

4.4 Configuration

Serial Ports

Configures the serial interface to the transceiver/transponder (This interface is usually connected to transceiver. If the user wants to configure a transponder he needs to unplug the transceiver cable and plug in the transponder cable.), GPS, compass, and external VRU if available. The user can also select a serial port to output the target positions and select a serial port for acoustic communication data. The user is advised to test all the interfaced devices before target tracking is started. See section 4.7 (Test) for more details.

Interrogation Rate

Configures how fast the target is interrogated. The minimum value is about 1.5 second depending on the tracking distance. The maximum value is 60 seconds.

Unit

Configures the unit (meter, foot or yard) used in plot and textual display.

Targets

Input the valid target addresses and names. The user could also choose the plotting target symbol on this window.

Warning: The user should avoid inputting addresses for a non-existing target. The software will time out on such non-existing target and cause delays in interrogating other valid targets. The user should make sure there is at least one valid target address is selected before clicking the “Start” button.

4.5 Input

Sound Speed

Inputs the sound speed near the TrackLink transceiver. The default value is 1500 meters/second. The value of sound speed will impact the accuracy of the range and bearing output.

For the TrackLink 5000LC system, the sound speed input is ignored due to this system’s relatively low accuracy.

4.6 Calibration

Load

Loads the calibration result generated by LinkQuest's automatic calibration software, TrackLink Calibrator. It will only be enabled when the user purchased the TrackLink Calibrator software.

Heading Offset

Inputs the heading offset caused by the installation. Please refer to section 5.3.1 and 5.3.2.

Pitch/Roll offset

Inputs the pitch/roll offsets caused by the installation. Please refer to section 5.3.3 Pitch and Roll Offset.

GPS Offset

Inputs the GPS offsets from the GPS antenna to the transceiver. The calibration procedures are discussed in section 5.3.4.

4.7 Test

Test Transceiver

When "Interrogate the Target" is selected, this function tests the serial communication with the transceiver, tracks the target once and determines whether the transceiver is functioning well in tracking. It returns error code when no target is available.

When "Interrogate the Target" is not selected, this function tests if the serial communication with the transceiver is good and if the rs232 line is clean and checks whether the transceiver is powered on properly.

Test GPS

Tests the GPS outputs for proper position data and frequency. The frequency of the data output is required to be no less than 2 Hz (twice every second).

The TrackLink Navigator assumes the GPS outputs NMEA data strings "GPGGA" or "GPGLL".

Warning: When GPS input is selected in “Serial Ports” setting, no valid position will be generated if no valid GPS information is available. In other words, the targets cannot be tracked in this case.

Test Compass

Tests the compass outputs for proper heading data and frequency. The frequency of the data output is required to be no less than 2 Hz (twice every second)

The TrackLink Navigator assumes the compass outputs NMEA data strings “HCHDM”, “HCHPR”, “HCHDT” or “HEHDT”.

Warning: When compass input is selected in “Serial Ports” setting, no valid position will be generated if no valid compass information is available. In other words, the targets cannot be tracked in this case.

Test VRU

Tests the VRU/motion sensor outputs. Currently the TSS format is supported. The format is as follows

:00ffeb 0004 -0505 –0316

Where the 3rd and 4th numbers are roll and pitch in 1/100 degree respectively.

Warning: When VRU/motion sensor input is selected in “Serial Ports” setting, no valid position will be generated if no valid VRU information is available. In other words, the targets cannot be tracked in this case

Noise

Measures the noise level at the transceiver. The user needs to power off and then power on the transceiver before running the noise test.

Debug 1

Configures the system to be in “Debug 1” mode. Tracking when “Debug 1” is activated will cause the system to record detailed debug information. Contact LinkQuest for details when the “Debug 1” feature is used to resolve problems in field deployment.

Debug 2

Configures the system to be in “Debug 2” mode and run the “Debug 2” test. It takes about five minutes to complete. Contact LinkQuest for details when the “Debug 2” feature is used to resolve problems in field deployment.

Debug 3

Configures the system to be in “Debug 3” mode and run the “Debug 3” test. It takes about 8 minutes to complete. Contact LinkQuest for details when the “Debug 3” feature is used to resolve problems in field deployment.

Debug 4

Configures the system to be in “Debug 4” mode and run the “Debug 4” test. It takes about 10 minutes to complete. Contact LinkQuest for details when the “Debug 4” feature is used to resolve problems in field deployment.

4.8 Transceiver

Test

When “Interrogate the Target” is selected, this function tests the serial communication with the transceiver, tracks the target once and determines whether the transceiver is functioning well in tracking. It returns error code when no target is available.

When “Interrogate the Target” is not selected, this function tests if the serial communication with the transceiver is good and if the rs232 line is clean and checks whether the transceiver is powered on properly.

Mode

Clicking on “Get” gets the transceiver’s operation mode. “Set” sets the selected mode for the transceiver. When the “Acoustic Interrogation” mode is selected, the transceiver interrogates all transponders acoustically. Therefore all transponders need to be set at “Transponder” mode. When the “Electrical Trigger” mode is selected, the transceiver interrogates the transponder with address 1 electrically while still interrogating transponders with all other addresses acoustically. In this case, the transponder with address 1 needs to be set at the “Responder” mode while all other transponders stay at the “Transponder” mode.

Power

Clicking on “Get” gets the transceiver’s operation power level. “Set” sets the selected power level for the transceiver.

Threshold

Clicking on “Get” gets the transceiver’s operation threshold. “Set” sets the selected threshold for the transceiver. This parameter determines the signal level that will be obtained by the system. Changing it will alter the system performance significantly. Please use caution to do so.

Version

Gets the firmware version of the transceiver.

Configuration Reset

Restore the original configuration of the transceiver. Please use caution when apply this command.

Note: Please always apply the "Get" function after sets a value to verify if the "Set" is successful.

Diagnostic Info

This function helps the user troubleshoot when the system is not working properly. Contact LinkQuest for details

4.9 Transponder

These functions are only valid for standard TrackLink transponders that support RS232 serial communication. The user should use the same serial port number used by the transceiver as configured in "Serial Ports" setting.

Test

Tests the serial communication with the transponder and determines whether the transponder is functioning.

Address

Retrieves and sets the addresses of the transponders. The address is from 1 to 8. Only address 1 could work in the electrical trigger mode.

Mode

Gets transponder mode and changes the transponder mode for testing purpose.

Clicking on "Get" gets the transponder mode. The transponder should be in the "Transponder" mode for the TrackLink system to function properly using acoustic interrogation. The "Responder" mode requires the transceiver to be configured to the "Electrical Trigger" mode.

When the transponder mode is set to the “Pinger” mode, the transponder will send out acoustic every second. This feature may be useful for troubleshooting the overall system.

Power

Clicking on “Get” gets the transponder’s operation power level. “Set” sets the selected power level for the transponder.

Threshold

Clicking on “Get” gets the transponder’s operation threshold. “Set” sets the selected threshold for the transponder. This parameter determines the signal level that will be detected by the system. Changing it will alter the system performance significantly. Please use caution to do so.

The user is highly recommended to use the default threshold value.

Version

Gets the firmware version of the transponder.

Wakeup Period

Gets and sets transponder wakeup period. Setting the wake up period to 0 will make the transponder wake up all the time. Therefore there will be no need to perform “Run->Wake Up” function for this transponder before the tracking starts.

Warning: Setting the wake up period to 0 will consume significantly more power.

Configuration Reset

Restores the original configuration of the transponder.

Diagnostic Info

This function helps the user troubleshoot when the system is not working properly. Contact LinkQuest for details.

4.10 Simulation

This function simulates the tracking output and display as if active tracking is in progress. In the case the user purchased a particular serial data output driver, this function will generate the serial output data and send the data to the serial port configured in Configuration->Serial Ports. The user can use this output to test the serial link interface to other software or instrument.

Run

Runs the tracking simulation.

Clear

Clears all the positions in the plots.

Stop

Stops the tracking simulation.

4.11 Help

Restore Database

This function is used to recover from errors caused by configuration information loss. If the user encounters some error message (for example error messages 13-18), and after restarting the program and rebooting the computer, the TrackLink Navigator software still does not work properly, this function can help the user to restore the database using the default configuration. After issuing "Restore Database", the user needs to close the TrackLink software and run it again. The user needs to reconfigure the system's serial ports and targets after Restore Database.

Software Version

The user gets the version of the TrackLink Navigator software.

Error Message

Shows the error message description.

4.12 Acoustic Communication

TrackLink system has optional integrated acoustic modem function to transfer 20 or 80 bytes data in each interrogation between the transceiver and transponder address 1. Higher data rate acoustic modem function is also available.

When the USBL transceiver interrogates the transponder. The user data will be piggybacked following the USBL interrogation signal. When the USBL transponder receives the interrogation, it will send a response and piggyback any acoustic modem data following it. By doing so, acoustic interference is avoided.

The user needs to configure a serial port for the acoustic communication. Please refer to section 4.4 Configuration. This port connects to a user's computer or device, which sends downlink data. The TrackLink software sends the data to the transceiver and the transceiver in turn sends it to the transponder address 1 when interrogating the transponder. The underwater device on the target sends uplink data to the transponder through a serial port. After the data are received at the transceiver, the data will be sent from the TrackLink Navigator software to user's computer through the designated serial port.

Note: The user should use a Null Modem converter between the TrackLink Navigator computer and the user's computer for input and output of acoustic modem data.

IMPORTANT: External power supply for transponder needs to be used for acoustic modem function due to the higher current requirement for continuous transmission. The peak current requirement for acoustic communication is 1.2 A at 27 volts.

For Type 8 (T8) acoustic communication, an integrated high speed UWM4010 acoustic modem is implemented for the uplink acoustic communication from the transponder to the transceiver. Refer to the user’s guide for UM4010 modem for interfacing issues. Please note the uplink data will be sent to a separate cable at the transceiver.

The acoustic communication relies on the success of the USBL interrogation. If the interrogation fails, the data will be lost.

4.13 Text Display

The positioning information is shown in textual format at the right hand corner of the software display. It shows the name of the target, the time of the interrogation, the longitude, latitude and heading of the ship. It also shows the X and Y positions of the target in the “Bow Reference” view, the longitude and latitude of the target, the (horizontal) range and slant range from the transceiver to the target, the bearing of the target relative to the ship (transceiver) and the depth of the target.

The unit (meter, foot or yard) used by X, Y, range, slant range and depth is shown at the left hand corner of the “Bow Reference” view plot.

The depth is a calculated value when depth sensor and acoustic communication options are not available. This value is less accurate at wider beamwidth. When no valid value can be calculated, an ‘*’ symbol is shown.

For all invalid or unavailable values, “*” symbols instead of numeric values will be shown.

The time-depth plot shows the target depth information for the past 100 interrogations. The depth information is calculated and may only be accurate within limited beam width.

The Acoustic Communication window displays up to 80 characters. The window will be updating when the next interrogation happens. A label above the window shows the communication direction, namely Downlink and Uplink. Downlink means the communication from the surface ship to the target. Uplink means the communication from the target to the surface ship.

The user should also pay close attention to the textual messages at the bottom right of the TrackLink Navigator software display for certain warning and error messages. **The most right cell shows the error code. Only "0" means tracking is successful.**

4.14 Tracking Log

The hp326.txt file saves the tracking history.

hp326.txt is in the working directory: C:\tracklink. The new data are appended to the end of the previous data. When the file size reaches 1.5Mbytes, it is copied to C:\tracklink\hist directory and renamed as hist +"current date and time".txt. Then the new data is saved to a newly created hp326.txt file.

Following is the format of the data string:

target_address error_code year month day hour minute second range bearing X Y Depth option_code shipLat shipLon option_code heading option_code TargetLat TargetLon SlangRange reserve option_code pitch roll reserve reserve reserve reserve reserve

If error code is not zero, ignore the position.

X, Y, Depth and Range are in meters. The SlantRange is in cm. Bearing, heading and pitch/roll are in degrees.

The user should check the files in the C:\tracklink\hist directory from time to time and remove old files if necessary.

4.15 Data Output Format

TrackLink software has three data output formats, namely TP2 compatible, GLL, and LXT compatible. The user may purchase one or more data output formats. Following are descriptions of these formats:

TP2 compatible output format:

1 11:08:51   0 229.4   199.7   -10.8   -9.3   199.2   0.0

Target number - Character 1
Time - Character 3-10
Reserved - Character 12-14
Target Bearing - Character 16-20
Slant Range - Character 22-28
X Offset - Character 30-37
Y Offset - Character 39-46
Z Offset - Character 48-54
Reserved - 56-63
Reserved - Character 65-66 (blank)
CR - 67
LF – 68

GLL output format:

$GPGLL,4916.45,N,12311.12,W,225444,A

4916.46,N   Latitude 49 deg. 16.45 min. North
12311.12,W   Longitude 123 deg. 11.12 min. West
225444   Fix taken at 22:54:44 UTC
A   Data valid
A followed by "CR" and "LF"

LXT compatible output format:

1 9999.9 359.9 89.9 359.9 9999.9 899.9 55
Target number
Slant range
Bearing
Depression angle
Reserved
Reserved
Reserved
Reserved

It is a fixed 41-character length not including CR, LF. This format is ended with "CR" and "LF".

Spaces are inserted for reserved fields.

5 Lab Test, Interfacing and Calibration

5.1 Lab Test

The user is recommended to perform a lab test of the system before operating the tracking system from the ship.

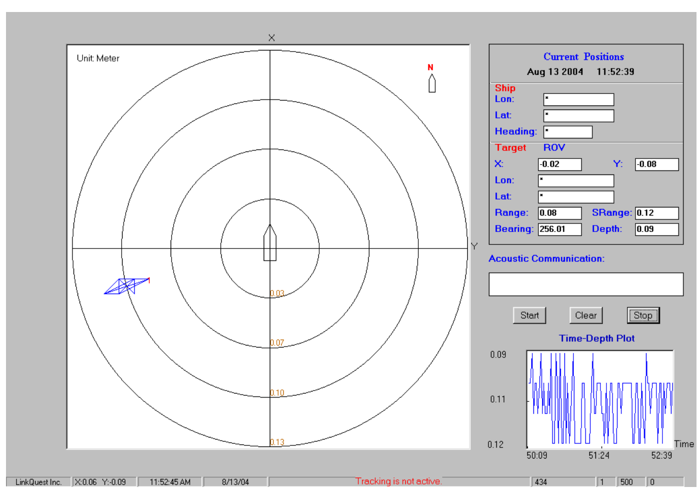
Place the transponder and transceiver into a small water tank or barrel, which can at least hold the transducer (black rubber part) of the transponder and the bottom (black urethane) part of the transceiver. It is recommended the size of the water tank or barrel is no less than 50 gallons.

<u>IMPORTANT: The distance between the top of the transponder and the bottom of the transceiver is recommended to be around 1 inch. The user should try to place the transponder and transceiver at center of the water tank, as far away from the tank walls, bottom and water surface as possible</u>.

Connect the transceiver to the computer and power on it with the transceiver cable. Power on the transponder using external power or the power-on plug if the transponder has internal battery. By using the TrackLink Navigator interface:

- Setup the target address. See 4.4 Configuration
- Configure the transceiver serial port. See 4.4 Configuration
- Wake up the transponder. See 4.2 Run.
- Test the transceiver. See 4.8 Transceiver.
- Run the tracking software. The software will display the target positions in the plots and show the textual positioning information.

Followings are the illustrations of TrackLink systems in testing in a small office waste can. The user is recommended to use a larger water tank or barrel for easier setup of the lab test.



Note: The pictures are for illustration purpose. The exact color and size of the system may vary.

Note: Occasional jumping of target positions on the display is possible due to the severe multipath conditions inside a small water tank. The user should adjust the positions of the transceiver and transponder to find the optimal setup for the test.

5.2 Interfacing

The TrackLink Navigator software interfaces to a transceiver, a GPS, a compass and a VRU through serial ports. It may also send out positioning data to a serial port. And send or receive acoustic communication data. Depending on the options (drivers) the user purchased from LinkQuest, up to 6 serial ports may be required.

If the user does not have existing multi-serial ports, an USB-serial port converter may be used to convert the USB port to support multiple serial ports.

If the user does not have existing multi-serial ports, an USB-serial port converter may be used to convert the USB port to support multiple serial ports.

IMPORTANT: <u>when a USB/RS232 converter is used to provide serial port support, the use needs to test the reliability of this serial port.</u> There are occasions computer crashes when such a serial port is used to interface to GPS, compass or motions sensor. When the problem occurs, the user is advised to use the original serial port of the computer (typically COM1) to interface to the external devices.

The interface to GPS and compass are optional. Without interfacing to GPS and compass, only the target's relative position to the ship (The Bow Referenced view) is displayed. The North Reference view is disabled.

After the user connects to the transceiver and other external devices through the serial ports, the user needs to go to "Configuration" on the menu bar and select "Serial Ports". In the pop-up window, enable an interface to a device by checking the box left of the device name before selecting the COM port and the baud rate corresponding for this particular device. If a device is not interfaced, the user needs to disable the device by un-checking the box left of the device.

After the configuration is completed, the user needs to go to "Test" on the menu bar and test each device before proceeding with normal tracking operations.

5.3 Calibration

The user can follow the simple procedures below to calibrate the TrackLink 5000 system. For users who purchased the TrackLink Calibrator software, he/she can load the result generated by the automatic calibration software by selecting the Calibration->Load Calibration function.

5.3.1 Heading Offset

Heading Offset is caused by the installation of the transceiver when it is not aligned with the bow of the vessel (see Figures 2a & 2b). The offset is used to adjust the target's bearing information (relative to the transceiver) to the target's actual heading based on the knowledge of the ship's heading information.

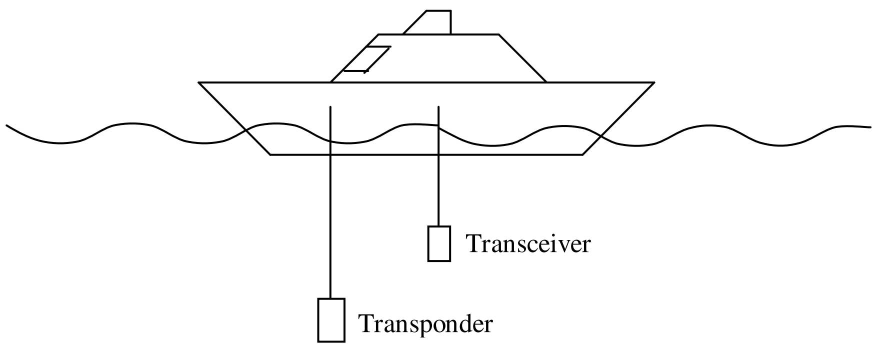


Figure 2a

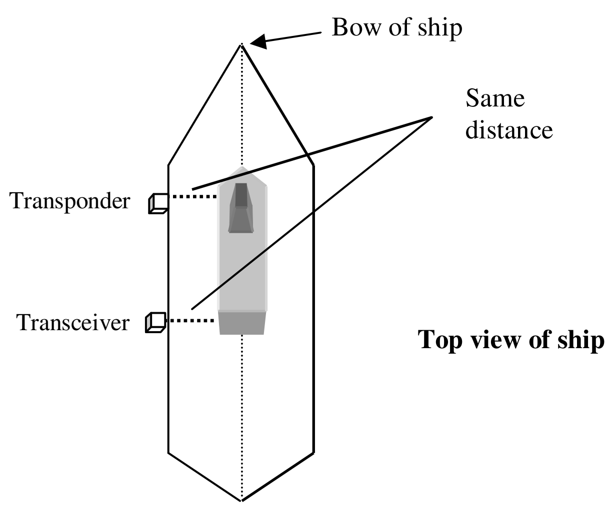


Figure 2b

5.3.2 Determination of a Heading Offset

With the TrackLink transceiver on, place a transponder directly in line with the transceiver off the bow of the vessel. Track the transponder with the TrackLink Navigator software. Note the relative bearing displayed on the software. This value should be centered about 0 degrees (i.e. between 358 to 2 degrees). Repeat the procedure until the relative bearing is within acceptable limits. If an offset exists substantially beyond these limits then enter a correction by selecting Calibration -> Heading Offset.

5.3.3 Pitch and Roll Offset

Place a pitch and roll sensor, a VRU or a motion sensor on the fixture (e.g. a steel pole) where the transceiver is installed. Make sure the sensor is leveled with the end-cap of the transceiver and +pitch points to the +X direction of the transceiver. Record the pitch and roll information from the sensor. Average the values if necessary. Enter the offset by selecting Calibration -> Pitch/Roll Offset.

If the transceiver is tilted to the stern direction, the pitch offset value is negative. Refer to Figure 3.

The user should also input the pitch offset if the transceiver is tilted to the stern intentionally to track a target, e.g. a towfish, behind the ship's stern.

If the transceiver is tilted to the port direction, the roll offset value is positive. Refer to Figure 4.

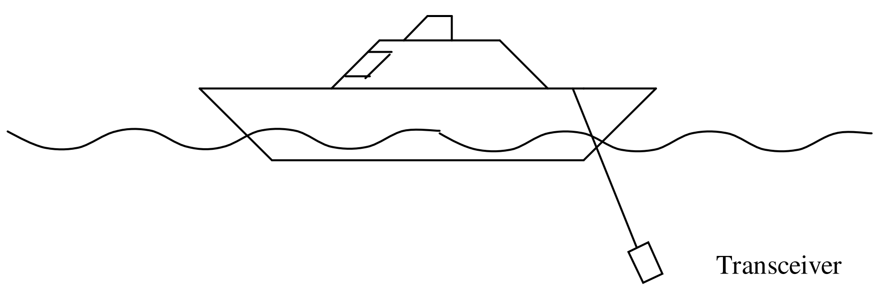
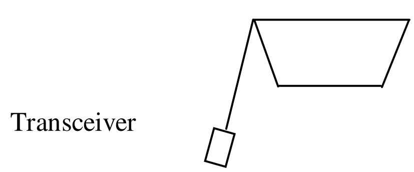


Figure 3



Ship Stern View

Figure 4

5.3.4 GPS Offset

The TrackLink Navigator software uses the positioning information from the GPS installed on the ship. When the GPS antenna is located far away from the transceiver, the offset from the transceiver to the GPS antenna needs to be compensated for. Enter the offset by selecting Calibration -> GPS Offset.

For example in Figure 5, the GPS X Offset is -5 meters while the GPS Y offset is -4 meters.

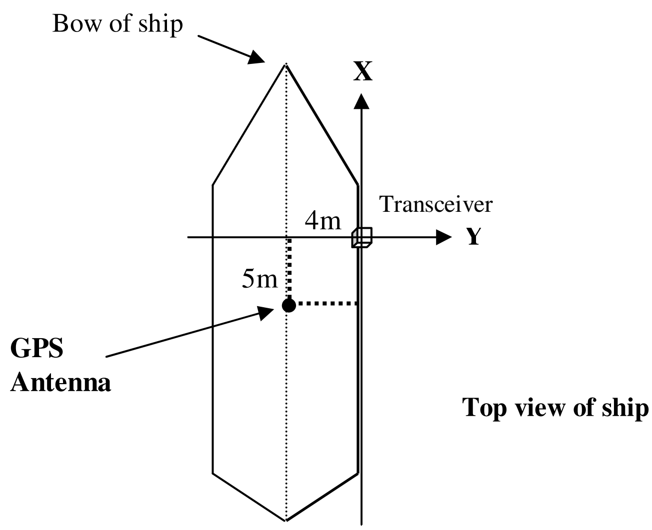


Figure 5

5.4 Operation

After the interfacing and calibration are completed, the user is ready to track targets with the TrackLink system.

LinkQuest's TrackLink transceiver uses a widebeam hydrophone array. It has the benefits of a directional receiver to better reject the ship's engine noise and other surface noise while gaining more flexibility in tracking targets as compared to a narrow-beam hydrophone array.

LinkQuest's TrackLink 5000 system (hydrophone array) is optimized to track targets within 120 degrees of beamwidth (+/- 60 degrees). If a target is between 120 to 140 degrees of beamwidth, it may be tracked with decreased accuracy and robustness. Although targets may be tracked beyond 120 degrees of beamwidth, it is highly recommended that the user DOES NOT plan to track targets beyond the 120-degree beamwidth (see Figure 6). If the transponder's beamwidth is smaller than transceiver's beamwidth or the acoustic transmitter of the transceiver is used, the effective beamwidth of the whole system is determined by the transponder's beamwidth or the acoustic transmitter's beamwidth (70 to 90 degrees), whichever is smaller.

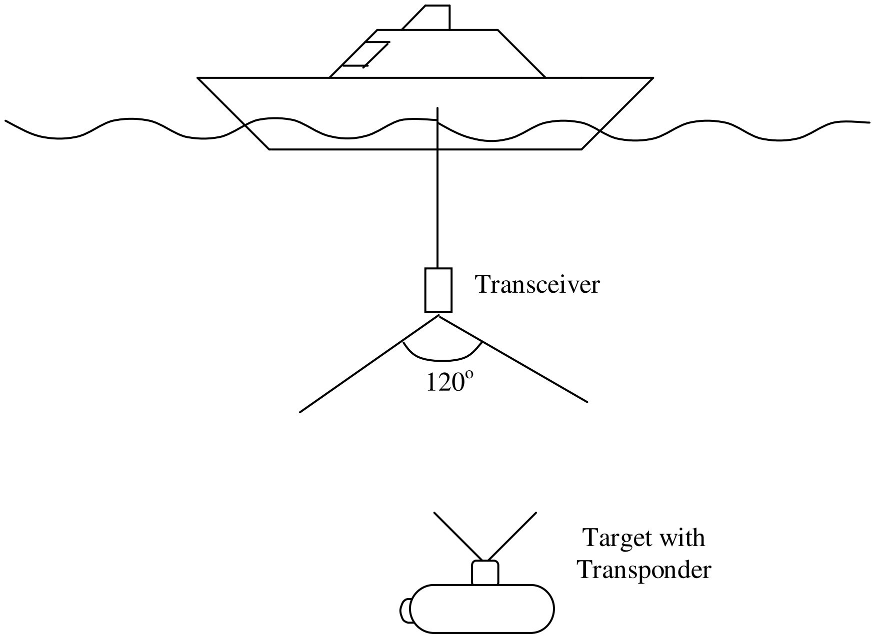


Figure 6

After powering on the systems, the user is advised to test all the devices such as the transceiver, GPS, and compass connected to the PC using the “Test” features on the TrackLink Navigator software menu bar. If all the tests passed, the user can start to track the targets.

The user should pay attention to the message box at the bottom of the Window. If a warning message appears instead of “Tracking is active …” message, the user should address this issue immediately. The software may display one or more of the following warning messages:

- "WARNING: Can't Communicate with Transceiver." The software cannot communicate with the transceiver. In this case, the serial link may be broken or the transceiver may be malfunctioned or powered off.

- “WARNING: No Valid GPS Data.” The software has not received valid data from the GPS. In this case, the GPS may be malfunctioned or the serial link between the PC and the GPS is broken. It is also possible that the GPS output of the valid data is not frequent enough. It is required that the output frequency be no less than 2 Hz.

- “WARNING: No Valid Compass Data.” The software has not received valid heading data from the compass. In this case, the compass may have malfunctioned or the serial link between the PC and the compass is broken. It is also possible that the compass has not provided data frequently enough. It is required that the output frequency be no less than 2 Hz.

- "WARNING: No Valid VRU Data." The software has not received valid pitch and roll data from the compass. In this case, the VRU may have malfunctioned or the serial link between the PC and the VRU is broken. It is also possible that the VRU has not provided data frequently enough. It is required that the output frequency be no less than 2 Hz. For better compensation of the pitch and roll of the ship, the output frequency is recommended to be higher than 30 Hz.

- "WARNING: No Valid Transceiver Data" The transceiver may have malfunctioned or lost the synchronization with the TrackLink Software. Power off the transceiver and try again.

- "WARNING: RS232 Link to Transceiver is NOT Clean!" The serial link between the transceiver and the computer is not clean. This is a very serious problem. It is usually caused by improper installation. The user may check the grounding of the computer and etc. This issue must be addressed before performing the tracking.

If the user selects a particular device in "Serial Ports", the tracking software will not be able to display a position if the selected device cannot output the required valid data.

Warning: The user should minimize the shock to the transceiver system installed from the ship. The shock limits for the transceiver system is defined in the Specification. When the ship moves at high speeds, the user should remove the transceiver from the steel pool so that the vibration caused by high-speed motion does not damage the transceiver.

6 Troubleshooting

The TrackLink Navigator software returns intuitive error messages and error codes when an error is detected. The details can be found in Appendix E.

The "Debug 1", "Debug 2", "Debug3" and "Debug4" features under "Test" on the menu bar provide various ways for the user to collect more detailed debug information. The user is advised to inquire more information from LinkQuest if further details are needed.

7 Customer Support

If technical problems with the instruments cannot be resolved using the procedures instructed in the troubleshooting section, the user should contact his/her local representative or contact LinkQuest directly in any of the following ways:

- Telephone: (858) 623-9900, 623-9916, 623-9919
- Fax: (858) 623-9918
- Email: support@link-quest.com
- Internet: http://www.link-quest.com
- Mail to: LinkQuest Inc.
  6749 Top Gun Street, Suite 100
  San Diego, CA 92121
  U.S.A.

Appendix A: TrackLink 5000 System Specifications

Positioning Accuracy:
TrackLink 5000LC: 3.0 degrees
TrackLink 5000MA: 1 degree
TrackLink 5000HA: 0.25 degree

Slant Range Accuracy: 0.30 meter

Targets Tracked: 8

Operating Frequency: 14.2 kHz – 19.8 kHz without acoustic modem option
12.75 kHz – 21.25 kHz with acoustic modem option

Transceiver operating Beamwidth: 120 degrees (hydrophone array)
70 to 90 degrees (acoustic transmitter)

Working range with ship noise: up to 5000 m with TN5015 transponders
up to 4000 m with TN5010 transponders

Maximum transceiver depth: up to 20 m

Input voltage: 46 to 48 Volts DC

Transmit mode power consumption: 40 Watts

Receive mode power consumption: 1.8 Watts

Peak Current Requirement: 2.5 amps at 48 Volts

RS-232 Configuration: 1 start bit, 1 stop bit, no parity bit, and no flow control

Hydrophone Array Dimension: 26 cm x 12.6 to 16 cm (d)
Acoustic Transmitter Dimension: 28 cm x 14 cm (d)

Hydrophone Array Weight in water: 2.3 kg
Acoustic Transmitter Weight in water: 3.6 kg

Operating temperature: -5 to 45 degrees
Storage temperature: -25 to 75 degrees

Optional High Speed Acoustic Modem Data Rate: up to 4800 baud

Shock: 30 g for the transceiver, 30 g for the transponders
Instruments contain shock sensor. Warranty is void if the sensor indicates more than 30 g force is applied to the instrument during usage or transit.

Appendix B: TrackLink 5000 Series Intelligent High Power Transponders

A wide range of transponders is available for the TrackLink acoustic tracking system. All the transponders use a state-of-the-art DSP to manage the power usage efficiently. The broadband Spread Spectrum technology used by the TrackLink system further decreases the power consumption. The DSP is programmed to stay in the standby mode most of time and wakes up to intercept acoustic signal periodically. After the transponder is awakened by the surface TrackLink transceiver, it transitions to the active mode and ready to respond to surface transceiver interrogation. The transponder will go to sleep after a prolonged period (1 hour) with no signal reception.

Each transponder has 8 configurable addresses stored in the flash memory and the user can conveniently use the RS232 interface to configure addresses and other parameters.

TN5010 is a high power directional transponder. TN5015 is a high power narrow beam (+/- 15 degree) directional transponder. Model A (e.g. TN5010A) is a compact transponder with no batteries. Model C is equipped with alkaline D cells for long term field use. . The TN5010 transponders have a remote transducer option. The letter “R” following the model number, e.g. TN5010CR, indicates the transponder has the remote transducer option.

The transponder needs to be powered externally during acoustic modem transmission. In this case, the peak current requirement is 1.2 A at 27 volts.

All transponders operate at frequency band from 14.2 kHz - 19.8 kHz without the acoustic modem option and 12.75 kHz - 21.25 kHz with the acoustic modem option

TN5010

Transmit Power: 100 Watts
Beamwidth: 60 degrees
Depth Rating: 4000 m
Optional Depth Rating: 2000 or 7000 m
Typical Range with Ship Noise: 4000 m

5010A:

Dimension: 43 cm x 15.2 cm (d)
Weight in Water: 5.9 kg
Weight out of Water: 11.8 kg
Input Voltage: 18 to 28 V (without acoustic communication)
20 to 28 V (with acoustic communication)
Peak Current: 0.6 A at 27 V (without acoustic communication)
1.2 A at 27 V (with acoustic communication)

5010C:

Dimension: 71 cm x 15.2 cm (d)
Battery Storage Time: 5 years
Battery Operation Time: 3 years
Active Responding Time:
8 x 100 hours
Weight in Water: 10.2 kg
Weight out of Water: 19.4 kg
Input Voltage: 26 to 28 V
Peak Current: 0.6 A at 27 V (without acoustic communication)
1.2 A at 27 V (with acoustic communication)

TN5015
Transmit Power: 1000 Watts
Beamwidth: 30 degrees
Depth Rating: 4000 m
Optional Depth Rating: 2000 or 7000 m
Typical Range with Ship Noise: 5000 m

5015A:
Dimension: 40 cm x 15.2 to 20 cm (d)
Weight in Water: 9.6 kg
Weight out of Water: 16.2 kg
Input Voltage: 18 - 28 V
Peak Current: 0.6 A

5015C:
Dimension: 82 cm x 15.2 to 20 cm (d)
Battery Storage Time: 5 years
Battery Operation Time: 3 years
Active Responding Time: 130 hours
Weight in Water: 16.2 kg
Weight out of Water: 28.4 kg
Input Voltage: 26 - 28 V
Peak Current: 0.6 A

Appendix C: Transceiver Cable and Connector Specification

The 8-conductor MICRO 8 SUBCONN connector used by the transceiver can be purchased from MacArtney Offshores, Inc. worldwide. They have offices in Denmark, Norway, Sweden, France and the United States. The part number of this connector is

MCBH8M (male connector)

The proper matching 8-conductor MICRO 8 SUBCONN cable connector can also be purchased from MacArtney Offshores, Inc. The part number of this matching cable connector is

MCIL8F (female cable connector)

The 8-conductor MICRO 8 SUBCONN transceiver cable should have shielding for proper grounding. The cable with the 8-conductor MICRO 8 SUBCONN connector can be ordered from LinkQuest Inc. For reliable RS-232 communication it is advised that the maximum cable be less than 20 meters. The user may choose to configure the other end of the cable (without the 8-conductor connector) to DB9, DB25 (PC connectors) or any other user preferences.

Table 1 provides the pin numbers and functions for the male connector on the transponder/transceiver (MCBH8M). The user should match the pin numbers on Table 1 with those in Figure 1.

Table 1: Male Connector Pin Assignment

male pin number	pin function	DB9 pin number	DB25 pin number
1	input data	3	2
2	output data	2	3
3	battery (+)	NA	NA
4	Ground	5	7
5	Reserved	NA	NA
6	Reserved	NA	NA
7	battery (-)	NA	NA
8	Trigger out	NA	NA

NA = not applicable

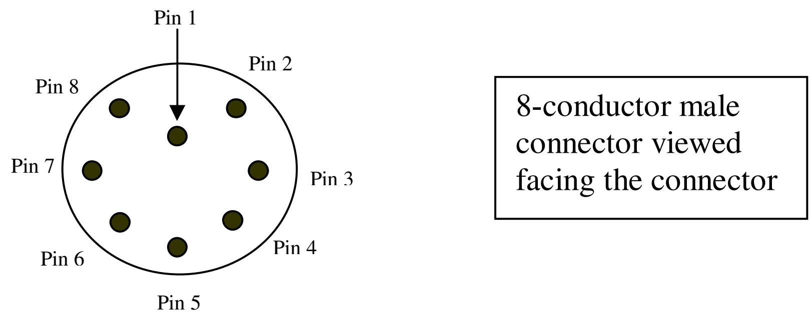


Figure 1

The trigger out pin sends out 3.3 volts logical signal in the “Electrical Trigger” mode. This signal can be send to user’s cable, e.g. ROV cable, to trigger the responder.

RS-422 Option

If the cable from the transceiver to the PC is relatively long and the EMI is relatively strong in the working environment, RS-422 may be the proper choice for connection as oppose to RS-232.

LinkQuest provide an RS-232/RS-422 converter as an option. The converter is used to connect a PC (or other RS-232 instruments) to the transceiver with the RS-422 option. Please refer to section 8.0 for the type and function of the transceiver’s connector.

The following figure is a plot of the converter.

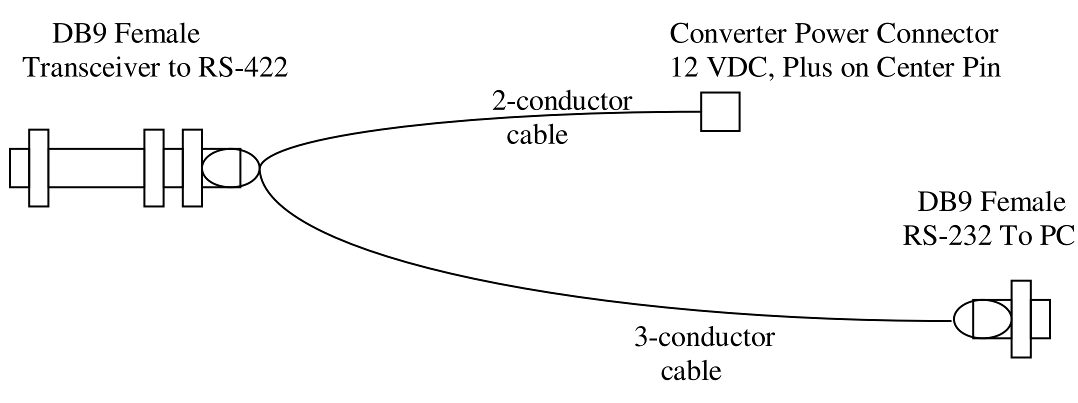


Figure 2

The power supply of the converter is 12 VDC. The current is less than 40 mA. The center pin of the power supply connector is plus.

Table 2 shows the pin assignment for the link between the RS-232 connector and the PC (i.e. this is compatible with the standard PC RS-232 connector assignment.). The 3-conductor cable carries the data to and from the transceiver as well as the ground connection.

Table 2: RS-232 connector to the PC

DB9 Pin Number	Pin function
1	NA
2	Data output from the transceiver
3	Command input to the transceiver
4	NA
5	Ground
6	NA
7	NA
8	NA
9	NA

NA = not applicable

The pin assignment between the transceiver's 8-conductor male connector and the RS-422 converter's pin assignment is provided in Table 3. Note that the RS-422 pin assignment is compatible with the RS-232 pin assignment (i.e. the cable with the RS-422 connector can also be used by the system with the RS-232 connector).

Table 3: Transceiver to RS-422 converter

Transceiver Pin Number	RS-422 Pin Function	DB9 Pin Number
1	RB	3
2	TA	2
3	Battery (+)	NA
4	Ground	5
5	TB	7
6	RA	8
7	Battery (-)	NA
8	Trigger out	NA
NA = not applicable		

****IMPORTANT****

If the RS-232/RS-422 communication line is disconnected or the power to the RS-232/RS-422 converter is disconnected while the transceiver is still powered on, junk data may be generated and transmitted to the underwater instruments. These junk data may be misinterpreted as a transceiver command. Therefore, ***it is very important to disconnect the transceiver power prior to disconnecting the RS-232, RS-422 or RS-232/RS-422 converter power.***

Appendix D: Transponder Cable and Connector Specification

The 8-conductor MICRO 8 SUBCONN connector used by the transponders can be purchased from MacArtney Offshores, Inc. worldwide. They have offices in Denmark, Norway, Sweden, France and the United States. The part number of this connector is

MCBH8M (male connector)

The proper matching 8-conductor MICRO 8 SUBCONN cable connector can also be purchased from MacArtney Offshores, Inc. The part number of this matching cable connector is

MCIL8F (female cable connector)

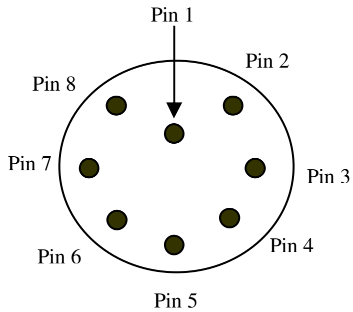
The 8-conductor MICRO 8 SUBCONN transceiver/transponder cable should have shielding for proper grounding. The cable with the 8-conductor MICRO 8 SUBCONN connector can be ordered from LinkQuest Inc. For reliable RS-232 communication it is advised that the maximum cable be less than 20 meters. The user may choose to configure the other end of the cable (without the 8-conductor connector) to DB9 or any other user preferences.

Table 1 provides the pin numbers and functions for the male connector on the transponder (MCBH8M). The user should match the pin numbers on Table 4 with those in Figure 1.

Table 1: Male Connector Pin Assignment

male pin number	pin function	DB9 pin number
1	input data	3
2	Output data	2
3	Battery (+)	NA
4	ground	5
5	reserved	NA
6	Trigger in	NA
7	battery (-)	NA
8	Power up	NA

NA = not applicable



8-conductor male connector viewed facing the connector

Figure 1

For all transponder models, pin 8 must be connected to pin 3 for proper transponder operation. For models with internal battery, this can be accomplished by plugging the power-up plug shipped with the transponder into the transponder male connector. If a custom cable (not provided by LinkQuest) is used, pin 8 and pin 3 need to be connected for the proper operation of the transponder.

The power-up plug should not be plugged into the transponder during storage. Otherwise the transponder will be powered on and the battery will be drained.

If an external 24-volt power supply is connected to pin 3 and pin 7, the external power supply will be used instead of the internal battery. In this case the power up pin (pin 8) also needs to be connected to pin 3 to power up the inside transponder circuit.

Responder Mode

The “trigger in” pin takes 3.3 volt or 5 volt logic signal with an input resistance larger than 2k ohms. The transponder, if configured at the “Responder” mode, will send out acoustic signal to the transceiver when the trigger signal is received.

Please note the “Responder” mode only works for address 1 and only when the transponder is externally powered. When the external power is cut off, the transponder will automatically switch to the “Transponder” mode and use internal power supply in about 20 seconds. If the external power supply is restored, the transponder will switch back to the “Responder” mode in about 20 seconds.

External Power Supply Connection Requirement

An external power supply connected to pin 3 can be used for the transponder. In this case the following two wiring requirements need to be satisfied to make sure that the transponder will always switch to internal battery power supply when the external power fails.

1. Make sure the power-up pin (pin 8) is connected to pin 3.
2. Connect a diode (SR206) in series with the external power supply before connecting it to pin 3 (refer to Figure 2).

Important: If the above diode is not used the internal power will be used when the voltage of the external power is lower than the battery voltage or the external power is disconnected. However if the external power is shorted to the ground the transponder will be powered off. The diode in series with the external power supply makes sure this will not happen.

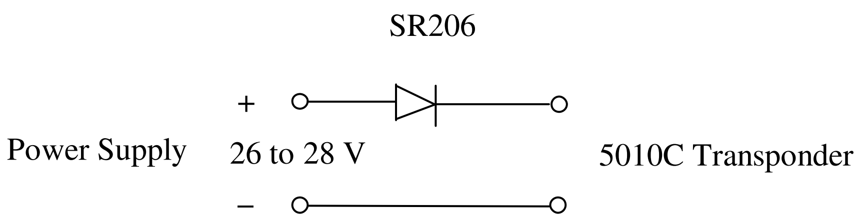


Figure2

Appendix E: Error Messages

Error 1: "RS232 has timed out."
The device has no valid RS232 data output to the computer. Use the "Test" feature of TrackLink Navigator software to see if the device functions properly.

Error 2: "RS232 input is lost."
The RS232 communication between the device and the computer is having problems. Check if the serial link between the device and the computer functions properly.

Error 3: "Transceiver has no responses."
Check if the transceiver is properly connected and powered on.

Error 4: "Conf table has no entry for GPS."
The user needs to specify GPS serial port on Configuration−>Serial Ports menu.

Error 5: "Conf table has no entry for Compass."
The user needs to specify Compass serial port on Configuration−>Serial Ports menu.

Error 6: "Conf table has no entry for Transceiver."
The user needs to specify Transceiver serial port on Configuration−>Serial Ports menu.

Error 7: "Conf table has no entry for VRU."
The user needs to specify VRU serial port on Configuration−>Serial Ports menu.

Error 8: "Transceiver has RS232 line error."
The user needs to check the RS232 connection between the transceiver and the computer.

Error 10: "Conf table has no entry for serial output."
The user needs to specify serial port for serial data output on Configuration−>Serial Ports menu.

Error 11: "Transceiver has timed out."
The transceiver has no valid RS232 data output to the computer. Use the "Test" feature of TrackLink Navigator to see if the transceiver functions properly.

Error 12: "Invalid Range."
It should occur occasionally. If this happens frequently, contact LinkQuest for support.

Error 13 - 18: "Cannot open file."
Internal file conflicts. If the user encounters one of the error messages 13-18 and other errors, and after restarting the program and rebooting the computer, the TranckLink Navigator software still does not work properly, restore database function can help the user to restore the database using the default configuration.

"Restore database" command is on the Help menu of the TrackLink Navigator software.

Error 19: "No valid GPS input."
The user needs to check the serial port setup for the GPS on Configuration−>Serial Ports menu. There is also possible that the GPS is not configured properly or does not function properly. Use the "Test" feature of TrackLink Navigator to check whether the GPS functions properly.

Error 20: "No valid Compass input."
The user needs to check the serial port setup for the compass on Configuration−>Serial Ports menu. The user needs to check the serial port setup for the Compass on Configuration−> Serial Ports menu. There is also possible that the Compass is not configured properly or does not function properly. Use the "Test" feature of TrackLink Navigator to check whether the Compass functions properly.

Error 21: "No valid VRU input."
The user needs to check the serial port setup (serial port and baud rate) for the VRU/Motion Sensor on Configuration->Serial Ports menu. The user needs to check the serial port setup for the VRU/Motion Sensor on Configuration−>Serial Ports menu. There is also possible that the VRU/Motion Sensor is not configured properly or does not function properly. Use the "Test" feature of TrackLink Navigator to see if the Compass functions properly.

Error 23: "Cannot open serial port for VRU."
Check if the port for VRU is valid.

Error 24: "No open serial port for transceiver."
Check if the port for transceiver is valid.

Error 25: "Cannot open serial port."
Check if the serial port is valid.

Error 26: "Cannot open serial port for GPS."
Check if the port for GPS is valid.

Error 27: "Cannot open serial port for Compass."
Check if the port for Compass is valid.

Error 28: "Transceiver error."
Cannot set transceiver parameter.

Error 29: "TrackLink Navigator error."
Save all the files in C:\tracklink directory and contact LinkQuest for support.

Error 30: "GPS driver is not available."
The GPS driver is not purchased. Contact LinkQuest for support.

Error 31: “Compass driver is not available.”
The compass driver is not purchased. Contact LinkQuest for support.

Error 32: “VRU driver is not available.”
The Motion Sensor/VRU driver is not purchased. Contact LinkQuest for support.

Error 33: “Serial output driver is not available.”
The serial output driver is not purchased. Contact LinkQuest for support.

Error 34: “Cannot open serial port for output.”
Check whether the port for serial data output is valid.

Error 37: “Cannot open file.”
See Error 13 -18.

Error 38 - 44: “Cannot set parameter.”
Contact LinkQuest for support.

Error 47: “Cannot open serial port for Acoustic Communication.”
Check if the serial port for acoustic communication is valid.

Error 48: “Acoustic Communication driver is not available.”
The acoustic communication driver is not purchased. Contact LinkQuest for support.

Error 49 - 50: “Cannot open file.”
See Error 13 -18.

Error 51: “Conf table has no entry for Acoustic Communication.”
Check “Configuration” -> “Serial Port” to see if the serial port for the acoustic communication is selected.

Error 52 - 53: “Cannot open file.”
See Error 13 -18.

Error 54 - 56: “Cannot set parameter.”
Contact LinkQuest for support.

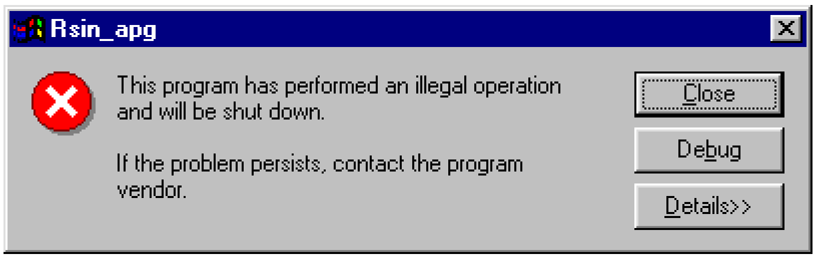
Error 58: “RS232 line is NOT CLEAN.”
Check the ground of the computer and etc. This problem has to be addressed before tracking any target.

Error greater than 64 and less than 100
Error codes those are greater than 64 and less than 100 are system errors. Power off transceiver for 10 seconds, then power on again. Please power off the transceiver first, and then close the warning window on the interface.

Error 127 - 128: "Motion sensor input error."
Input data from VRU/motion sensor are incorrect.

Error 180 - 182: "Motion sensor input error."
Input data from VRU/motion sensor are incorrect.

Error messages related to difficulty in opening a serial port (COM port) and error messages similar to the following:



It is very likely that one or more serial ports are not released after the TrackLink software or other user software exits. Please reboot the computer if restarting the TrackLink Navigator software does not address the problem.

Appendix F: Transponder Battery Core Replacement

1. Introduction

The replacement of the battery core involves three main parts — the anodized aluminum transponder housing, the battery pack and the battery core inside the battery pack. The battery core is replaceable and may be purchased directly through LinkQuest Inc.

To prevent water leakage into the pressure housing when replacing the battery core, the user should follow the procedures below for removal and re-installation with extra caution. Since the procedures for re-installation is similar to the procedures for removal, the user should take good notes during the removal process to help with the re-installation process.

Note: The pictures are for illustration purpose. The exact color and size of the system may vary.

2. Supplies and tools needed for battery core replacement

Please make sure the following supplies and tools are readily available:

- New battery core (refer to specification)
- Voltmeter
- Wooden chisel
- A clean piece of paper (letter size or greater)
- Transparent tape
- Plastic ties for securing wires
- Marker (e.g. small pieces of colored tape)
- Two flat stands of equal thickness for placing the battery pack vertically (e.g. books).
- Pliers, wrenches and hammer

3. Determining the condition of the old and new battery cores

Before replacing the battery core, the customer should check the voltage of the battery core inside the housing to determine the condition of the battery core. The nominal voltage of the ABP16D battery core used in TN5010C transponder is 24 volts.

1) Power on the transponder using the 2-meter testing cable (do not use the power on plug) by connecting "power on" pin to power positive. (Usually these two are connected by the manufacturer.) Open the TrackLink software and go to "Transponder" -> "Mode". Set the transponder to "Responder" mode. Using a

voltmeter (it is better to use a digital meter) to measure the voltage of the old battery core from the red and black power connectors.

2) If the voltage falls below 16 volts, the battery core needs to be replaced in order for the transponder to function properly.

3) Before replacing the old battery core, measure the voltage of the new battery core. For the new battery core purchased from LinkQuest (refer to specification), the positive and negative leads of the 2-pin connector are depicted in Figure 1. The new battery core should register a voltage of at least 24 volts for ABP16D (the new battery cores may register a voltage that is 1 or 2 volts above the nominal voltage).

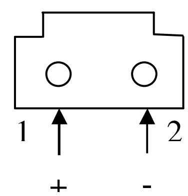


Figure 1: Diagram of the 2-pin connector viewed facing the connector

4. Procedures for removing the TN5010 existing battery core

The end cap of the battery pack contains two O-rings. These O-rings must be protected from contamination or damage at all times during the replacement procedures.

1) Place the transponder securely on a flat surface with ample work area.

2) Before removing the screws on the end cap, prepare a clean sheet of paper with an 8cm hole in the middle. This paper will be used to protect the two O-rings from contamination while the battery core is being replaced.

3) Remove the 6 screws that hold the end cap and retain the screws, nuts and washers for later use. Be sure to make a note of the order of the nut and the washers on both sides of the screw. The order of the washers is critical to proper operation of the transponder.

4) After the 6 screws are removed, carefully open (open slightly) the end cap using the wooden chisel and a hammer (refer to figure 3). Please be sure never use a metal screwdriver. Be careful not to stick the wooden chisel more than 1cm into the end cap so as not to damage the O-rings sitting inside the end cap. The user should open the end cap by working evenly on all sides of the end cap.

5) Before pulling out the end cap use a colored marking tape to align the hole on the end cap with the hole on the bottom of the housing (please refer to figure 4). If the holes on the end cap are not properly aligned with the holes on the bottom of the housing, the user will not be able to re-install the battery pack properly.

6) Using both hands, pull the end plate 4cm out of the housing (see figure 4). It is important that the user does not rotate the end plate relative to the housing while the end plate is being pulled out.

7) Locate the 14-pin ribbon cable within the 4cm gap between the end plate and the housing (see figure 4). If the 14-pin ribbon cable is located in the bottom portion, the user may rotate the housing and the end cap until the 14-pin ribbon cable appears on the top for easier access. It is important that the end cap and the housing be rotated simultaneously to avoid any damage.

8) Remove the transparent tape that holds the 14-pin ribbon cable to the battery pack.

9) Pull the 14-pin connector from its socket (see figure 5). Make sure to hold both side of the 14-pin connector before disconnecting the connector.

10) Using transparent tape affix the disconnected 14-pin connector onto the housing as shown in figure 6. Make sure the 14-pin connector is affixed just above the long screw of the battery pack; otherwise damage will occur during re-installation.

11) Cover the O-ring, which is on the cap, with the transparent tape.

12) Pull out the rest of the battery pack without rotating the end cap. The battery pack is long so make sure to use both hands to support both ends of the battery pack while its being pulled out.

13) Place the removed battery pack vertically on two stands with the end cap on the bottom (see figure 7). Make sure the 8-conductor connector underneath the end cap is not touching the table.

14) Remove the plastic tie that holds the 2 wires to long screw of the battery pack and disconnect the 2-pin connector from its socket at the bottom of the battery core (see figure 8).

15) Remove the four nuts on the top of the battery pack and remove the circular plate that holds the battery core.

16) Remove the battery core from the battery pack. Make a note of the location of the 2 wires on the battery core relative to the battery pack.

5. Procedures for installing the TN5010 battery core

After the old battery core is removed from the battery pack we are now ready to install the new battery core in the battery pack and re-install the battery pack into the housing.

1) Slide the new battery core onto the battery pack making sure that the 2-wires are at the same position as they were when the old battery core was on the battery pack.

2) Re-connect the 2-wire connector to its socket at the bottom of the battery core.

3) Secure the 2-wire leads to the long screw of the battery pack using a plastic tie.

4) Replace the circular metal plate on the top of the four long screws and replace the four nuts that hold the circular metal plate. It is important that the 4 nuts are not over tightened (see figure 9).

5) Replace the tape around the 2-pin connector and 14-pin ribbon cable.

6) Remove the tape on the housing cap O-ring.

7) Examine the two O-rings on the end cap. **The O-rings must be completely free of contaminants or damage.** If either O-rings is contaminated or damaged, the O-rings must be cleaned or replaced by a qualified professional. It is better to place some silicon grease on the O-rings for sealant purpose. LinkQuest uses DOW CORNING Compound 111.

8) Hold the battery pack securely with both hands and start inserting the battery pack into the housing**. Make sure the holes on the end cap are properly aligned with the holes on the bottom of the housing using the colored markers placed earlier.**

9) Do not insert the battery pack all the way into the housing. Leave 4cm gap between the end cap and the housing (see figure 4).

10) Reconnect the 14-pin connector to its socket (see figure 10) and secure the ribbon cable flat onto the battery pack using transparent tape. The ribbon cable must not bulge up above the surface of the battery pack otherwise the ribbon cable will obstruct with the final re-installation.

11) Measure the voltage from the bottom of the end plate using procedures in section 3. Make sure the voltage is correct (refer to specification).

12) There are two copper springs (see figure 4) on each side of the end plate. These copper springs are used for grounding and must have a good electrical contact

with the inside wall of the housing. Adjust the copper springs so that they are neither too far outward (which obstruct with the re-installation of the battery pack into the housing) nor too far inward (which causes poor or no electrical contact with the inside wall of the housing).

13) It is better to tape a small packet of desiccant on the battery core.

14) Push the battery pack all the way into the housing while making sure the big O-ring on the end cap is securely fitted inside the O-ring groove (see figure 11). Also make sure that the holes on the end cap are perfected aligned with the holes on the housing. The user may need to give the end cap a firm push to insert the last centimeter of space between the end cap and the housing (due to the tight fit of the bore O-ring).

15) Replace the six screws along with the nuts and washers that were set-aside during the removal process. **The washers and the nuts must be placed in the correct order as they were before removal.** If the user forgets the positions of the washers, use the positions of the washers on the other end of the housing as a point of reference.

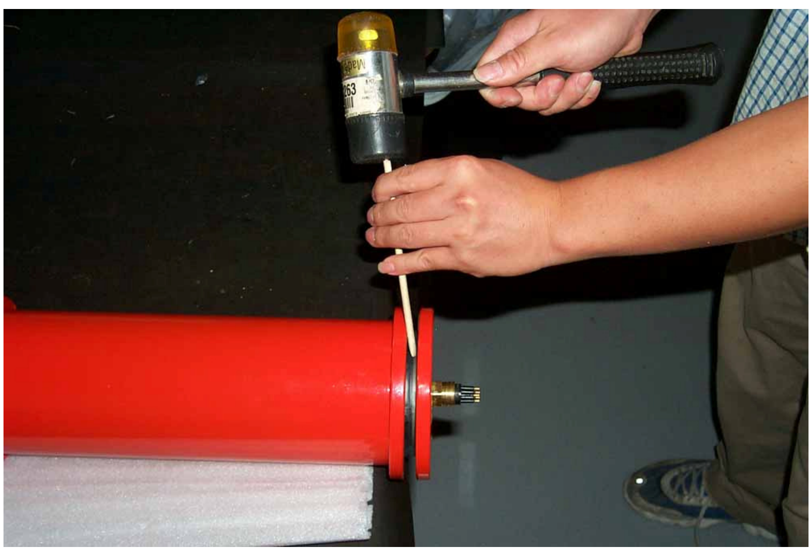


Figure 3

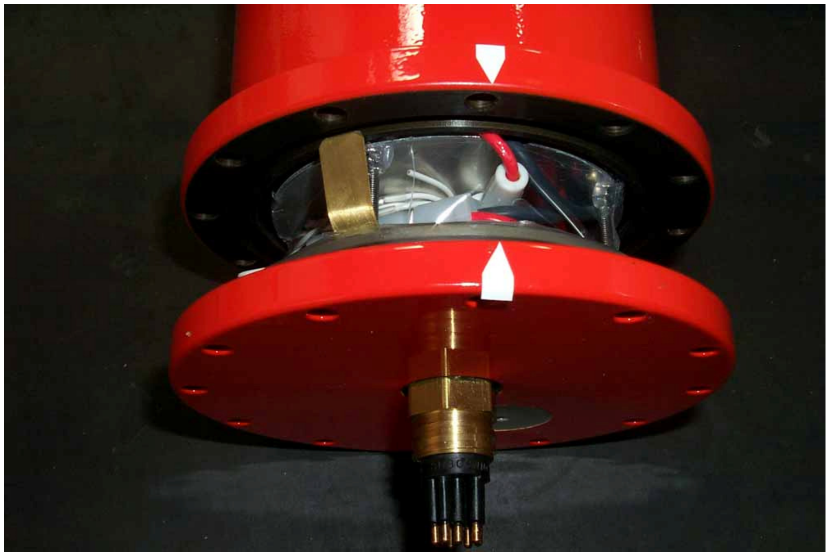


Figure 4

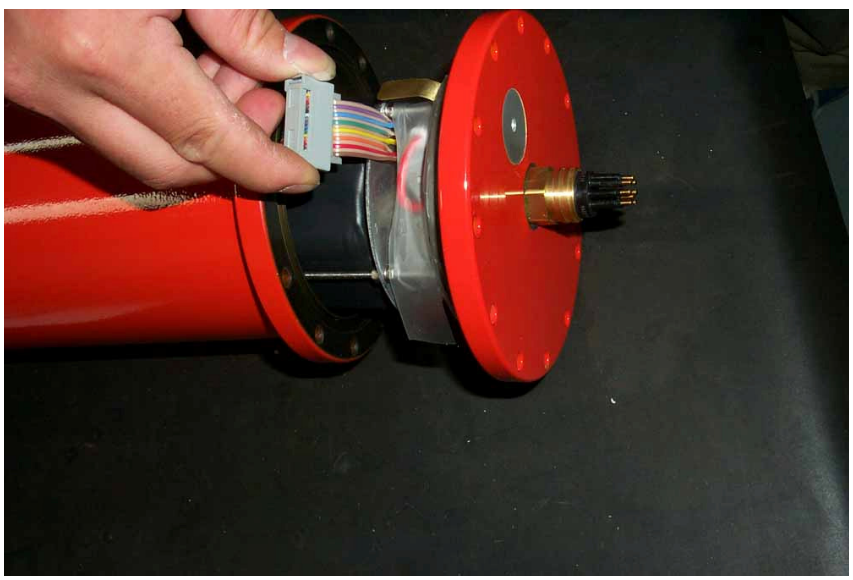


Figure 5

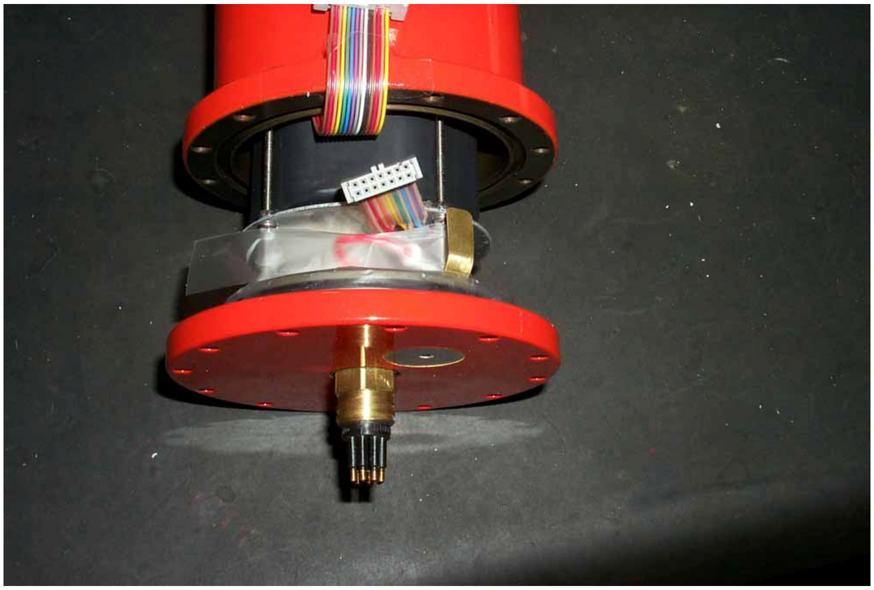


Figure 6

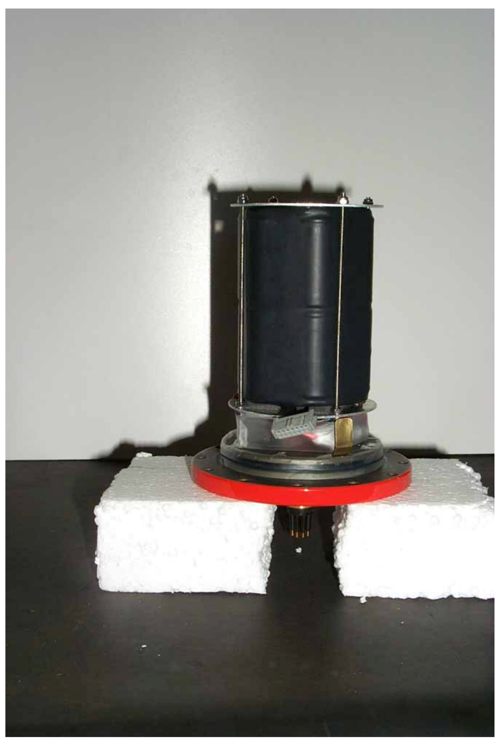


Figure 7

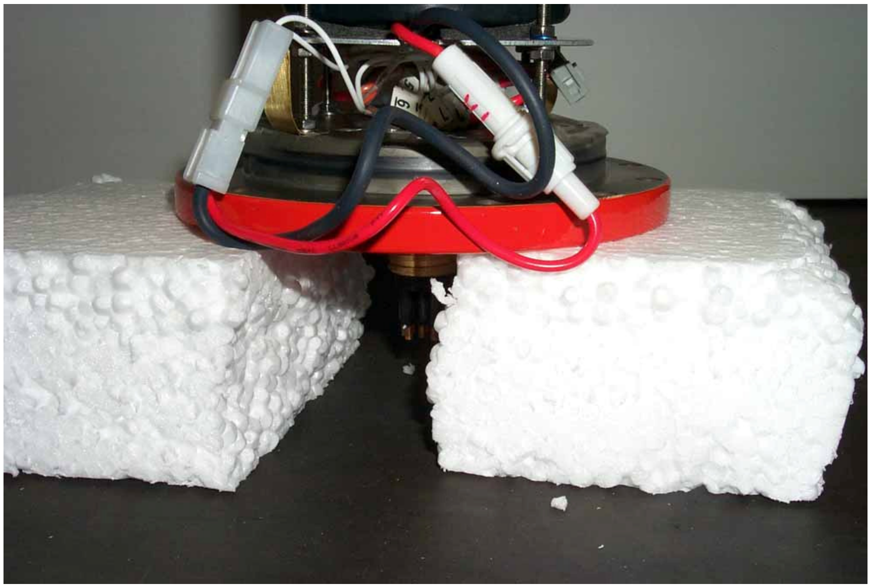


Figure 8

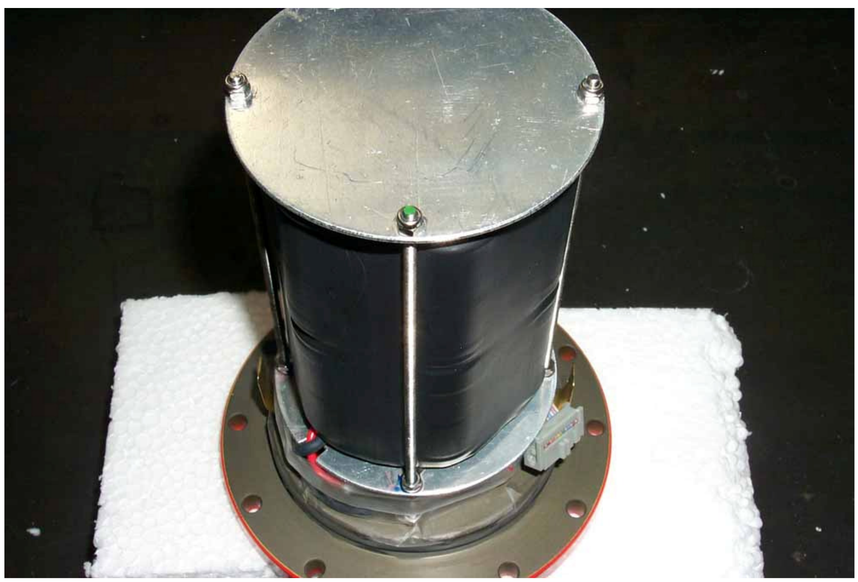


Figure 9

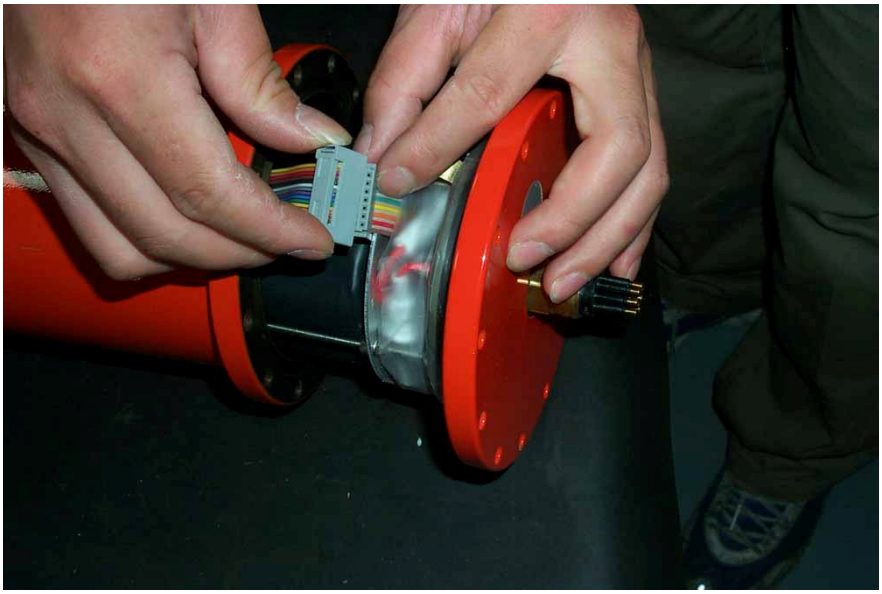


Figure 10

Figure 11

6. Specifications

Nominal Voltage: ABP16D: 24V (24 ~ 25.6V)

Bore O-Ring: Parker O-Ring Part # 2-237

Surface O-Ring: Parker O-Ring Part # 2-248

Appendix G: Make Your Own ABP16D Battery Pack

The user may choose not to buy the battery pack from LinkQuest and instead of make his/her own battery pack using 16 “D” size alkaline batteries. The schematic diagram for the battery core is depicted in Figure 1.

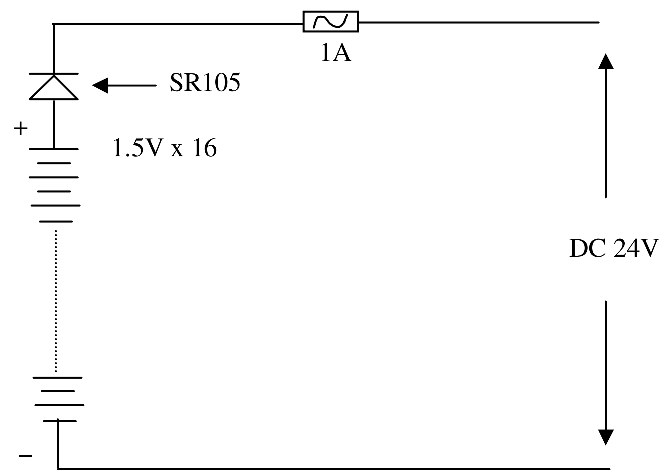


Figure 1

16 serially connected D size batteries are packaged as four layers and four columns as depicted in Figure 2. A diode (SR105) and a 1A fuse are in serial with the battery positive output.

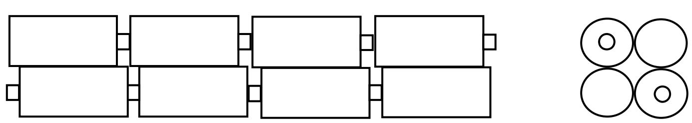
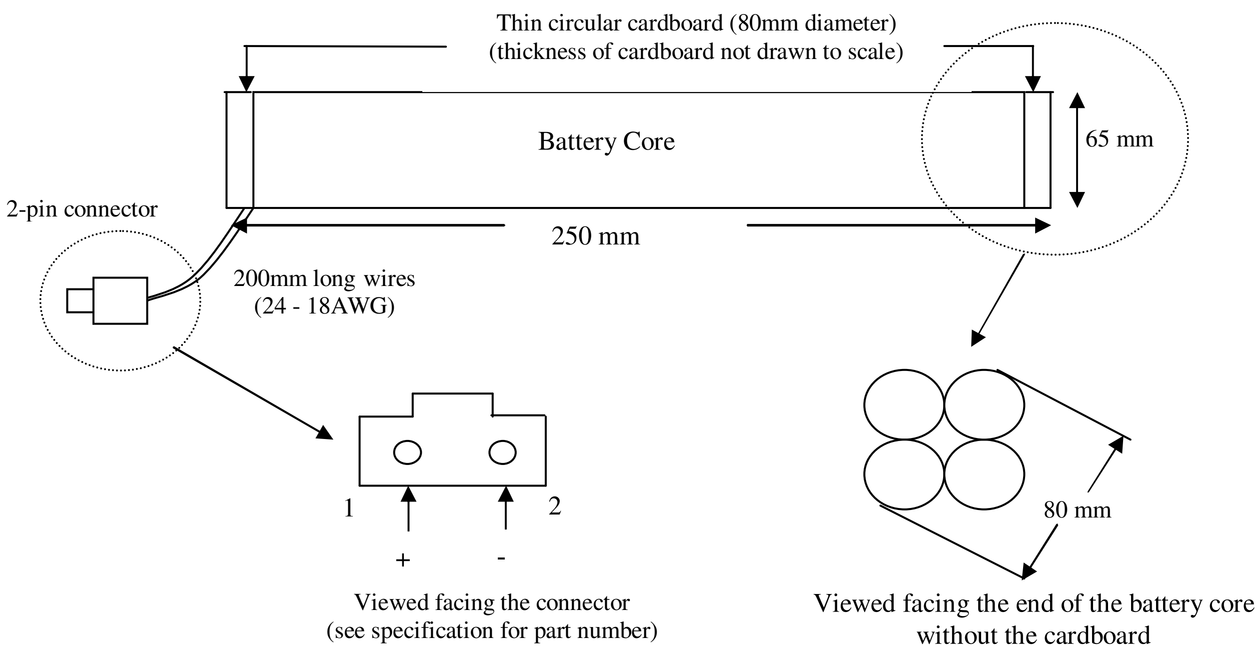


Figure 2

The negative and positive leads should be connected to 200mm long wires (24-18AWG) ended with the specified 2-pin connector (see Figure 3). The two ends of the battery pack should be secured with circular cardboards as in Figure 3. Once the wires are all connected and the batteries are properly arranged, wrap the outside of the battery core using a “heat shrink tubing”. Note that the “heat shrink tubing” is slightly longer than the length of the battery core for proper fitting. Make certain that the finished battery core has the proper dimension, 80 x 250; otherwise the battery core will not fit properly.



Thin circular cardboard (80mm diameter)
(thickness of cardboard not drawn to scale)
Battery Core
65 mm
2-pin connector
250 mm
200mm long wires
(24 - 18AWG)
1
2
+
-
80 mm
Viewed facing the connector
(see specification for part number)
Viewed facing the end of the battery core
without the cardboard

Figure 3

Limited Warranty

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The limited warranty does not cover any shipping charges and customs fees associated with the repair.

To obtain warranty service the warranty holder must first obtain a return invoice number from LinkQuest Inc. by calling the phone number listed below. The product must be returned with insurance and shipping prepaid, to LinkQuest Inc. at the address below, in its original packaging or a suitable equivalent, along with a written description of the problem.

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TrackLink System DSP Firmware & PC Tracking Software

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